CONE5112/w Exam Preparation Question Bank

Computer Networks 1B · Prescribed: Kurose & Ross (2022), *Computer Networking: A Top-Down Approach*, 8th ed.

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How to use this question bank

The summative exam is a 2-hour, closed-book, 100-mark paper covering all six learning units, with theory questions and scenario-based application questions. Each unit below offers three question types, mapped to the unit's learning outcomes (LO) and module outcomes (MO1–MO4).

Type	Marks	What it tests	Suggested time
A. Short answer	2–6	Define, list, describe, distinguish	2–6 min
B. Calculation / problem	4–10	Apply formulas and algorithms	5–12 min
C. Long / scenario	10–20	Explain, analyse, compare, evaluate, recommend	12–25 min

Rule of thumb: about 1 mark per 1.2 minutes, and one well-developed point per mark.

Task words used in the paper

- **Define / State** – a precise one-line meaning, no elaboration.
- **Describe** – what it is and how it works, in order.
- **Explain** – how and why; link cause and effect.
- **Compare / Distinguish** – similarities and differences on the same criteria; a table earns marks.
- **Analyse** – break a scenario into parts and show how each part affects the outcome.
- **Evaluate** – weigh strengths and limitations, then give a justified judgement.
- **Recommend** – choose, justify against the scenario, and note trade-offs.

Learning Unit 1: Computer Networks, the Internet, Wireless and Mobile Networks

Covers Kurose & Ross Chapters 1 and 7 (LO1–LO10).

A. Short-answer questions

1. Define the term "protocol" and give two examples from different layers. (3)
2. Describe the Internet as a "network of networks", referring to access ISPs, regional ISPs, tier-1 ISPs, IXPs and content-provider networks. (6)
3. Distinguish between packet switching and circuit switching under the headings: resource reservation, efficiency and suitability for bursty traffic. (6)
4. Name and briefly describe the four sources of nodal delay. (4)
5. Explain what is meant by encapsulation, using a message sent from an application down to the physical medium. (5)
6. List the five layers of the Internet protocol stack and state one service each provides. (5)
7. Describe three characteristics that distinguish a wireless link from a wired link. (3)
8. Compare infrastructure mode and ad hoc mode in IEEE 802.11 networks. (4)
9. Explain the difference between a handover (handoff) and roaming in a cellular network. (4)
10. State two advantages of 5G over 4G LTE. (2)

B. Calculation and problem questions

1. A 1,500-byte packet is sent over a 10 Mbps link that is 2,000 km long. Propagation speed is 2 × 10^8 m/s. Calculate (a) the transmission delay, (b) the propagation delay and (c) the total delay, ignoring processing and queuing. (6)
2. A file of 4 million bits is sent from Host A to Host B across three links with rates R1 = 500 kbps, R2 = 2 Mbps and R3 = 1 Mbps. (a) What is the throughput of the file transfer? (b) Roughly how long will the transfer take? (4)
3. A link carries traffic with average packet length L = 1,000 bits, link rate R = 1 Mbps and average packet arrival rate a = 800 packets/s. Calculate the traffic intensity and comment on the expected queuing delay. (4)
4. Using CDMA, a sender uses the code (1, −1, 1, −1) and transmits the data bit 1. Show the encoded output, and show how the receiver decodes it. (5)

C. Long and scenario questions

1. **Scenario – Rural clinic connectivity.** A provincial health department plans to connect 40 rural clinics in KwaZulu-Natal to a central patient-records system. Options under review are DSL over existing copper, fixed wireless, 4G LTE and fibre (FTTH). (a) Compare these four access technologies on bandwidth, cost, reliability and coverage. (8) (b) Recommend one option for clinics more than 30 km from the nearest exchange and justify your choice. (4) (c) Explain which delay components would most affect real-time video consultations and how they could be reduced. (6) **(18 marks)**
2. Discuss the layered architecture model. In your answer, explain the advantages and disadvantages of layering, and show how encapsulation supports interoperability between devices from different vendors. (12)
3. Analyse the architecture of a 4G LTE network, explaining the roles of the UE, eNodeB, MME, HSS, S-GW and P-GW, and describe what happens when a user moves between cells during a call. (15)
4. Evaluate Mobile IP as a solution to device mobility. Explain home agent, foreign agent, care-of address and triangle routing, and assess its limitations in practical deployments. (12)

Learning Unit 2: The Application Layer

Covers Kurose & Ross Chapter 2 (LO1–LO7).

A. Short-answer questions

1. Distinguish between client-server and peer-to-peer architectures, giving one example application of each. (4)
2. List four transport services an application might require, and give an application that needs each. (4)
3. Explain the difference between non-persistent and persistent HTTP connections. (4)
4. Describe the four components of a cookie-based system. (4)
5. Explain how a web cache (proxy server) reduces response time and access-link traffic. (4)
6. Describe the hierarchy of DNS servers: root, TLD and authoritative. (6)
7. Distinguish between iterative and recursive DNS queries. (4)
8. Explain the purpose of the conditional GET and the If-Modified-Since header. (3)
9. State two "enter deep" and "bring home" CDN server-placement strategies, describing one advantage of each. (4)

10. Explain why TCP socket programming requires a welcoming socket and a connection socket on the server side. (4)

B. Calculation and problem questions

1. A web page has one base HTML file and 8 referenced images. RTT = 50 ms; ignore transmission time. Calculate total response time with (a) non-persistent HTTP and no parallel connections, (b) persistent HTTP without pipelining, (c) persistent HTTP with pipelining. (6)
2. An institution has a 15 Mbps access link. Average object size is 1 Mb, and the request rate is 14 requests/s. (a) Calculate access-link utilisation. (b) If a web cache with a 40% hit rate is installed, calculate the new utilisation and comment on the effect on delay. (6)
3. A file of F = 15 Gb is distributed to N = 100 peers. Server upload rate us = 30 Mbps, each peer′s download rate d = 2 Mbps, each peer′s upload rate u = 700 kbps. Calculate the minimum distribution time for (a) client-server and (b) P2P. (8)

C. Long and scenario questions

1. **Scenario – Online learning portal.** A private college′s student portal slows down sharply during exam weeks, when 8,000 students stream lecture recordings and download study material at once. (a) Analyse the likely causes of poor performance at the application layer. (6) (b) Evaluate how web caching, HTTP/2 and a CDN would each improve performance. (9) (c) Recommend an approach and explain how DASH would improve the streaming experience for students on slow mobile connections. (5) **(20 marks)**
2. Describe step by step what happens at the application layer, from DNS resolution to the final HTTP response, when a user types www.iie.ac.za into a browser. (12)
3. Analyse how client-server and P2P communication models differ in scalability, performance and resource distribution. Use the distribution-time formulas to support your answer. (12)
4. Explain the principles of TCP socket programming by describing the sequence of calls made by a TCP client and server. Include pseudocode or a labelled diagram. (10)

Learning Unit 3: The Transport Layer

Covers Kurose & Ross Chapter 3 (LO1–LO7).

A. Short-answer questions

1. Explain the role of the transport layer and how it differs from the network layer. (4)

2. Distinguish between connectionless and connection-oriented demultiplexing. (4)
3. Give three reasons an application developer might choose UDP over TCP. (3)
4. Describe the four fields in the UDP segment header. (4)
5. Explain why sequence numbers and timers are needed in a reliable data transfer protocol. (4)
6. Describe the TCP three-way handshake, naming the flags set in each segment. (6)
7. Distinguish between flow control and congestion control. (4)
8. Describe the three phases of TCP Reno congestion control: slow start, congestion avoidance and fast recovery. (6)
9. Explain what is meant by TCP fairness, and why parallel TCP connections and UDP traffic undermine it. (4)
10. Explain how TCP estimates the retransmission timeout using EstimatedRTT and DevRTT. (4)

B. Calculation and problem questions

1. Compute the 1s-complement checksum of these three 16-bit words: 0110011001100000, 0101010101010101, 1000111100001100. Show how the receiver detects an error. (6)
2. Hosts A and B are 30 ms apart (RTT = 30 ms). The link is 1 Gbps and packets are 1,500 bytes. (a) Calculate sender utilisation for stop-and-wait. (b) What window size (in packets) is needed for 95% utilisation with a pipelined protocol? (6)
3. A TCP sender's cwnd starts at 1 MSS with ssthresh = 16 MSS. Draw or tabulate cwnd for transmission rounds 1 to 12, assuming a triple duplicate ACK occurs at the end of round 8 (TCP Reno). (8)
4. EstimatedRTT = 100 ms, DevRTT = 5 ms, $\alpha = 0.125$, $\beta = 0.25$. A new SampleRTT of 120 ms is measured. Calculate the new EstimatedRTT, DevRTT and TimeoutInterval. (6)
5. Host A sends two segments to Host B with sequence numbers 92 and 100. The first segment is lost. (a) What acknowledgement does B send after the second segment? (b) What happens when A's timer expires? (4)

C. Long and scenario questions

1. **Scenario – Banking app and live sports streaming.** A telecoms company hosts two services on the same data centre link: a mobile banking app and a live sports-streaming service. (a) Recommend a transport protocol for each service and justify your choice. (6) (b) During a major match, the link becomes congested and banking transactions time out. Analyse the causes and costs of this congestion. (6) (c) Evaluate how TCP congestion control and ECN would respond, and discuss the fairness implications of UDP-based streaming sharing the link. (8) **(20 marks)**

2. Compare Go-Back-N and Selective Repeat under the headings: sender window, receiver behaviour, buffering, acknowledgements, retransmission and efficiency on a lossy link. Include a timeline diagram of one lost packet for each. (14)
3. Explain how TCP provides reliable data transfer over an unreliable network layer. Refer to segment structure, sequence and acknowledgement numbers, cumulative ACKs, timers, fast retransmit and flow control. (15)
4. Discuss the historical evolution of transport-layer protocols, from early TCP (Tahoe) to Reno, CUBIC, and QUIC, and evaluate future trends. (12)

Learning Unit 4: The Network Layer – Data Plane

Covers Kurose & Ross Chapter 4 (LO1–LO7).

A. Short-answer questions

1. Distinguish between forwarding and routing, and between the data plane and control plane. (4)
2. Describe the four components of a generic router architecture. (4)
3. Explain where queuing can occur in a router (input and output ports) and describe head-of-line (HOL) blocking. (5)
4. Compare FIFO, priority queuing and weighted fair queuing. (6)
5. Explain the difference between the best-effort service model and a guaranteed service model. (4)
6. Describe how DHCP allocates an IP address to a new host, naming the four DHCP messages. (4)
7. Explain how NAT works and give two advantages and two disadvantages. (6)
8. Describe three changes in the IPv6 datagram format compared with IPv4. (3)
9. Explain tunnelling as an IPv4-to-IPv6 transition mechanism. (4)
10. Define a middlebox and give three examples. (4)
11. Contrast traditional per-router control with SDN's logically centralised control. (4)

B. Calculation and problem questions

1. A company is allocated 192.168.10.0/24. It needs four subnets with 50, 25, 12 and 6 hosts. Using VLSM, give each subnet's network address, prefix, usable range and broadcast address. (10)
2. A router has this forwarding table. Determine the output interface for each destination address using longest-prefix matching. (6)

Prefix	Interface
200.23.16.0/21	0
200.23.24.0/24	1
200.23.24.0/21	2
Otherwise	3

- (a) 200.23.22.161 (b) 200.23.24.10 (c) 200.23.30.5 (d) 200.23.40.1

3. A 4,000-byte datagram (20-byte header) must cross a link with an MTU of 1,500 bytes. Calculate the number of fragments and, for each, the data length, offset and flag values. (6)
4. For the IP address 172.16.45.200/20, determine the network address, broadcast address, subnet mask and number of usable hosts. (5)

C. Long and scenario questions

1. **Scenario – Campus network redesign.** A university campus in Durban runs four departments (Administration 120 hosts, IT 60, Library 25, Labs 200) on one flat network using the private range 10.20.0.0/16. Students report slow access and staff are concerned about security. (a) Design an addressing plan for the four departments, showing prefixes and ranges. (8) (b) Explain how NAT and DHCP would be used in the design. (6) (c) Recommend whether to start a dual-stack IPv6 deployment, and justify your answer. (6) **(20 marks)**
2. Explain the inside of a router: input port processing, switching fabrics (memory, bus, interconnection network), output port processing, buffering and packet scheduling. Evaluate how each affects router performance. (15)
3. Compare traditional destination-based forwarding with generalised forwarding (match-plus-action) in SDN. Use OpenFlow examples to show how one flow table can act as a router, switch, firewall and NAT. (12)
4. Discuss the transition from IPv4 to IPv6. Explain why it is needed, the main differences in datagram structure, and evaluate dual-stack and tunnelling approaches. (12)

Learning Unit 5: The Network Layer – Control Plane

Covers Kurose & Ross Chapter 5 (LO1–LO7).

A. Short-answer questions

1. Distinguish between per-router control and logically centralised (SDN) control in the control plane. (4)
2. Distinguish between global (centralised) and decentralised routing algorithms, and between static and dynamic algorithms. (4)
3. State the Bellman-Ford equation and explain each term. (4)
4. Explain poisoned reverse and the problem it addresses. (4)
5. Describe four features of OSPF, including hierarchical areas. (4)
6. Distinguish between eBGP and iBGP sessions. (4)
7. Explain hot-potato routing. (3)
8. Describe the BGP route-selection rules, in order. (5)
9. Explain IP anycast and how it differs from unicast and multicast, giving DNS root servers as an example. (5)
10. Describe the four components of a network management framework: managing server, managed device, MIB data and agent. (4)
11. Distinguish between SNMP and NETCONF/YANG approaches to network management. (4)
12. Explain how the ping and traceroute utilities use ICMP. (4)

B. Calculation and problem questions

1. Use Dijkstra's algorithm to compute the least-cost paths from node u to all other nodes in this network. Show a table of each step and draw the resulting forwarding table for u. (10)

Link	Cost
u–v	2
u–w	5
u–x	1
v–w	3
v–x	2
x–w	3
x–y	1
w–y	1

Link	Cost
w–z	5
y–z	2

2. In a network of three nodes, x–y = 2, y–z = 1 and x–z = 7. Using the distance-vector algorithm, show the distance tables at x, y and z after initialisation and after each exchange until convergence. (8)
3. In a separate three-node network where x–y = 4, y–z = 1 and x–z = 50, the cost of link x–y increases from 4 to 60. Explain why routing loops occur and how many iterations are needed before convergence. (6)

C. Long and scenario questions

1. **Scenario – Multi-homed ISP.** A South African ISP (AS 65010) connects to two upstream providers and peers with a local content provider at an Internet Exchange Point in Johannesburg. (a) Explain how eBGP and iBGP allow routers within AS 65010 to learn routes to external prefixes. (6) (b) The ISP wants customer traffic to a streaming service to exit through the IXP rather than an upstream provider. Explain which BGP attributes and policies would achieve this. (6) (c) Explain why the ISP would not want to carry transit traffic between its two upstream providers, and how routing policy prevents this. (4) (d) Recommend an intra-AS protocol for the ISP's internal network and justify your choice. (4) **(20 marks)**
2. Compare link-state and distance-vector routing algorithms on message complexity, speed of convergence, robustness and typical use cases. (12)
3. Explain why the Internet uses a hierarchical routing structure with separate intra-AS and inter-AS protocols. Refer to scale, administrative autonomy and policy. (10)
4. Discuss the role of network management in maintaining network health. Explain SNMP operation (get, set, trap), the MIB, and evaluate why NETCONF and YANG are increasingly preferred. (12)

Learning Unit 6: The Link Layer and LANs

Covers Kurose & Ross Chapter 6 (LO1–LO7).

A. Short-answer questions

1. List and describe four services the link layer may provide. (4)
2. Distinguish between broadcast links and point-to-point links, with an example of each. (4)

3. Compare parity checks, the Internet checksum and CRC on detection strength and where each is used. (6)
4. Describe the three broad classes of multiple access protocols, giving one example of each. (6)
5. Explain how CSMA/CD works, including binary exponential backoff. (5)
6. Describe the fields of an Ethernet frame. (6)
7. Explain how a switch learns and builds its switch table (self-learning), and what happens when a destination is unknown. (5)
8. Compare switches and routers under: layer of operation, addressing, plug-and-play, and traffic isolation. (6)
9. Explain three problems with flat LANs that VLANs solve. (3)
10. Describe a hierarchical data centre architecture (top-of-rack, tier-1 and tier-2 switches, border router) and explain how it improves performance. (6)
11. Explain the roles of DHCP, UDP, IP, ARP, Ethernet and DNS when a laptop joins a network and requests a web page. (6)

B. Calculation and problem questions

1. Using even two-dimensional parity, construct the parity matrix for the data 1110 0110 1001 0101 (four rows of four bits). Show how a single-bit error in row 2, column 3 is detected and corrected. (6)
2. Data D = 101110 and generator G = 1001. Calculate the CRC bits R and the bits transmitted. Show that the receiver detects no error. (8)
3. Using slotted ALOHA, with N nodes each transmitting with probability p = 1/N, explain why the maximum efficiency approaches 1/e as N becomes large. (4)
4. Host A (IP 111.111.111.111) on subnet 1 sends a datagram to Host B (IP 222.222.222.222) on subnet 2, through router R. Describe the ARP queries and the source and destination MAC and IP addresses of each frame. (8)

C. Long and scenario questions

1. **Scenario – Corporate office LAN.** A financial services firm in Cape Town has 300 staff across Finance, HR, Sales and Guest Wi-Fi on one switched network. Broadcast traffic is high and auditors have flagged that guests can see Finance traffic. (a) Analyse the problems with the current flat design. (5) (b) Design a VLAN-based solution, explaining access ports, trunk ports, 802.1Q tagging and inter-VLAN routing. (10) (c) Explain how switch port security and segmentation would harden the network. (5) **(20 marks)**

2. Describe and compare the three classes of multiple access protocols. Evaluate which is best suited to (a) a heavily loaded network and (b) a lightly loaded network, and justify your answer. (15)
3. "A day in the life of a web page request." Trace every protocol used, from the moment a student's laptop connects to a campus Ethernet port to the moment a web page is displayed. Cover DHCP, ARP, DNS, TCP, HTTP and routing. (20)
4. Explain how the link layer interacts with the network and physical layers to provide end-to-end communication, using the example of a frame crossing two switches and a router. (10)

Mock examination: integrated case study (100 marks, 2 hours)

This paper mirrors the summative format: one scenario, questions across all six learning units. Answer all questions. Closed book.

Case study – Ubuntu Logistics (Pty) Ltd

Ubuntu Logistics is a courier company headquartered in Durban, with a regional depot in Pietermaritzburg and 150 drivers on the road. Head office has 180 staff in Operations, Finance, IT and Customer Service. The depot has 40 staff. Drivers use a mobile app over 4G/5G to receive delivery routes and upload proof-of-delivery photos. Customers track parcels on a public website that peaks at 20,000 visitors per hour during the festive season. The company has been allocated 10.50.0.0/16 for internal use and one public /29 block. Currently the head office runs a single flat LAN, and both sites connect to one ISP. Management wants better performance, security and resilience before December.

Question 1 – Network fundamentals and mobility (LU1) (15 marks)

1. Explain why the Internet is described as a "network of networks", and show where Ubuntu Logistics, its ISP and its customers sit in this hierarchy. (5)
2. A 5 MB proof-of-delivery photo is uploaded over a 4G link with an upload rate of 10 Mbps. Calculate the transmission delay and explain one other delay component that affects the driver's experience. (4)
3. Explain how a driver moving between cell towers keeps an active session with the company's server in a 4G/5G network. (6)

Question 2 – Application layer (LU2) (15 marks)

1. Explain the DNS process when a customer enters track.ubuntulogistics.co.za for the first time. (6)
2. Evaluate how a CDN and web caching would help the tracking website cope with festive-season traffic. (6)

3. State whether the tracking website should use HTTP/1.1 or HTTP/2 and justify your answer. (3)

Question 3 – Transport layer (LU3) (15 marks)

1. Recommend TCP or UDP for (a) the photo upload and (b) live GPS location updates from drivers sent every 2 seconds. Justify each. (6)
2. Explain how TCP congestion control would respond if the head-office ISP link became congested on Black Friday. (6)
3. Explain how TCP flow control prevents a slow server from being overwhelmed by uploads. (3)

Question 4 – Network layer data plane (LU4) (20 marks)

1. Using VLSM from 10.50.0.0/16, design subnets for Operations (90 hosts), Customer Service (60), Finance (20), IT (12) and the depot (40). Show network address, prefix and usable range for each. (10)
2. Explain how NAT would allow 220 staff to share the public /29 block. (5)
3. Explain how a middlebox such as a firewall could be implemented using SDN match-plus-action rules. (5)

Question 5 – Network layer control plane (LU5) (15 marks)

1. Recommend an intra-AS routing protocol for linking head office and the depot, and explain how it calculates routes. (6)
2. Management wants to add a second ISP for resilience. Explain the role of BGP in a multi-homed connection. (5)
3. Explain how the IT team would use ICMP and SNMP to monitor and troubleshoot the network. (4)

Question 6 – Link layer and LANs (LU6) (20 marks)

1. Analyse the problems of the current flat LAN at head office. (4)
2. Design a VLAN solution for the four departments and guest Wi-Fi. Explain access and trunk ports, 802.1Q and inter-VLAN routing. (10)
3. Explain how ARP and switch self-learning allow a Finance PC to send a frame to the default gateway. (6)

Memo: answers and marking pointers

Answers below cover every calculation. Long questions are marked by the rubric: one mark per accurate, relevant point, with extra credit for application to the scenario.

Calculation answers

Question	Answer
LU1 C1	Transmission = 12,000 bits ÷ 10 Mbps = 1.2 ms; propagation = 2 × 10^6 m ÷ 2 × 10^8 m/s = 10 ms; total = 11.2 ms
LU1 C2	Throughput = min(R1, R2, R3) = 500 kbps; time ≈ 4 × 10^6 ÷ 5 × 10^5 = 8 s
LU1 C3	La/R = (1,000 × 800) ÷ 10^6 = 0.8; delay grows sharply as intensity approaches 1
LU1 C4	Output = (1, −1, 1, −1); receiver computes (1·1 + 1·1 + 1·1 + 1·1) ÷ 4 = 1, so bit = 1
LU2 C1	(a) 2 RTT + 8 × 2 RTT = 18 RTT = 900 ms (b) 2 RTT + 8 RTT = 10 RTT = 500 ms (c) 2 RTT + 1 RTT = 3 RTT = 150 ms
LU2 C2	(a) 14 Mbps ÷ 15 Mbps = 93%, queuing delay very high (b) 0.6 × 14 = 8.4 Mbps → 56%, delay drops to near LAN levels for hits
LU2 C3	F = 15,000 Mb. (a) Client-server: max(NF/us = 50,000 s, F/d = 7,500 s) = 50,000 s (b) P2P: max(F/us = 500 s, F/d = 7,500 s, NF/(us + Nu) = 1,500,000 ÷ 100 = 15,000 s) = 15,000 s
LU3 C1	Sum of words 1 and 2 = 1011101110110101; add word 3 = 1 0100101011000001; wrap carry = 0100101011000010; checksum = 1011010100111101. Receiver sums all four words; any 0 in the result signals an error
LU3 C2	L/R = 12,000 ÷ 10^9 = 0.012 ms; (a) U = 0.012 ÷ 30.012 ≈ 0.0004 (0.04%) (b) N ≈ 0.95 × 30.012 ÷ 0.012 ≈ 2,376 packets
LU3 C3	Rounds 1–8: 1, 2, 4, 8, 16, 17, 18, 19. Triple duplicate ACK: ssthresh = 9. Rounds 9–12: 9, 10, 11, 12 (accept 12, 13, 14, 15 if the +3 MSS fast-recovery inflation is shown)

Question	Answer
LU3 C4	EstimatedRTT = 0.875 × 100 + 0.125 × 120 = 102.5 ms; DevRTT = 0.75 × 5 + 0.25 × |120 − 100| = 8.75 ms; Timeout = 102.5 + 4 × 8.75 = 137.5 ms (accept 8.125 ms and 135 ms if the updated estimate is used)
LU3 C5	(a) ACK 92 (cumulative; B still awaits byte 92) (b) A retransmits the segment with sequence number 92 only
LU4 C1	50 hosts: 192.168.10.0/26, .1–.62, bcast .63 · 25 hosts: .64/27, .65–.94, bcast .95 · 12 hosts: .96/28, .97–.110, bcast .111 · 6 hosts: .112/29, .113–.118, bcast .119
LU4 C2	(a) 0 (b) 1, longest match /24 (c) 2 (d) 3
LU4 C3	3,980 data bytes; 1,480 per fragment. Fragment 1: 1,480 bytes, offset 0, MF = 1 · Fragment 2: 1,480 bytes, offset 185, MF = 1 · Fragment 3: 1,020 bytes, offset 370, MF = 0
LU4 C4	Mask 255.255.240.0; network 172.16.32.0; broadcast 172.16.47.255; 4,094 usable hosts
LU5 C1	Least costs from u: x = 1 (u–x), v = 2 (u–v), y = 2 (u–x–y), w = 3 (u–x–y–w), z = 4 (u–x–y–z). Forwarding table: v via (u, v); x, y, w, z via (u, x)
LU5 C2	After one exchange: Dx(z) = min(7, 2 + 1) = 3 via y; Dz(x) = 3 via y; y unchanged. Second exchange brings no change: converged
LU5 C3	Without poisoned reverse, y routes to x via z and z via y; costs climb by 2 per exchange. 44 iterations until z prefers its direct 50-cost link
LU6 C1	Row parities: 1, 0, 0, 0; column parities: 0, 1, 0, 0; corner bit 1. A flipped bit at row 2, column 3 makes row 2 and column 3 parities fail; their intersection locates and corrects it
LU6 C2	Append three 0s: 101110000 ÷ 1001 gives remainder R = 011; transmit 101110011; receiver divides by 1001, remainder 0
LU6 C3	Efficiency = Np(1 − p)^(N−1); with p = 1/N, (1 − 1/N)^(N−1) → 1/e ≈ 0.37 as N → ∞

Question	Answer
LU6 C4	A ARPs for R′s left-interface MAC; frame 1: src MAC A, dst MAC R-left, src IP A, dst IP B. R ARPs for B; frame 2: src MAC R-right, dst MAC B, IPs unchanged

Mock examination pointers

- **Q1.2:** 5 MB = 40 Mb ÷ 10 Mbps = 4 s transmission delay; propagation or queuing delay at the cell tower adds to it.
- **Q4.1 (VLSM, largest first):** Operations 10.50.0.0/25 (.1–.126); Customer Service 10.50.0.128/26 (.129–.190); Depot 10.50.0.192/26 (.193–.254); Finance 10.50.1.0/27 (.1–.30); IT 10.50.1.32/28 (.33–.46).
- **Q3.1:** TCP for photo upload (integrity); UDP for GPS updates (timeliness, a lost update is replaced 2 s later).
- **Q5.1:** OSPF expected (link-state, Dijkstra, fast convergence, areas); credit RIP only if limits are noted.
- **Q6.2:** Award marks for a VLAN table (ID, name, subnet), trunk between switches, router-on-a-stick or Layer 3 switch, and guest isolation.